

Lesson Cards

One accessible curriculum-review card per lesson, built from current app data plus inventory notes and rubric scores.

1 / 28

What Is an LLM?

A learned next-token prediction system

Stage: architecture Priority: low Rubric avg: 4.4/5

DEFINITION
A large language model is a learned system that turns current context into probabilities for the next token.

DURABLE VS TEMPORARY
The weights were changed during training or fine-tuning. During ordinary inference, hidden states are temporary and weights normally stay fixed.

RELATIONSHIP
Next, we compare the older rule-based dream of AI with the deep-learning approach that made LLMs work.

BRAIN LIMIT
A person has a body, goals, lived experience, and awareness. The model has learned weights and temporary computations.

CURRENT EXERCISE
pick-next-token: Pick the Next Token

REWRITE NOTES
Risk: An LLM is a conscious reader or a database.
Missing: An LLM does not read a prompt and write a whole answer at once. During inference, the current context enters the model, learned weights shape internal hidden states, the model scores possible next tokens, one token is chosen, and that token is appended before the next run. Training built the weights earlier; inference uses them.

CORE EXPLANATION
An LLM does not read a prompt and write a whole answer at once. During inference, the current context enters the model, learned weights shape internal hidden states, the model scores possible next tokens, one token is chosen, and that token is appended before the next run. Training built the weights earlier; inference uses them.

PROMPT VS RESPONSE
Prompt tokens are given to the model. Response tokens are generated one at a time.

METAPHOR
A vast autocomplete engine with learned structure.

MISCONCEPTION
An LLM is a conscious reader or a database.

CHECKPOINT
Which description is most accurate? Correct: A learned prediction system. Incorrect: A conscious reader; A hand-written rulebook; A search engine that knows every file.

ILLUSTRATION NEEDED
Current visual aid: llm-overview. One-page prompt lifecycle: prompt enters, fixed weights process, next token sampled, response grows.

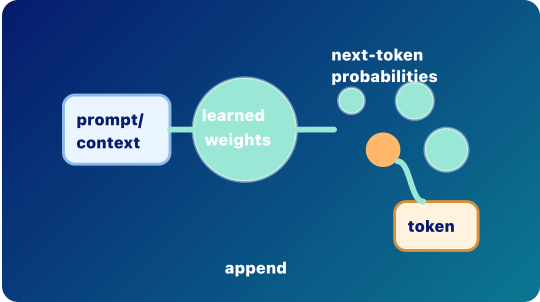
WHERE IT HAPPENS
This card describes the whole model lifecycle, but especially normal inference: current context to next-token probabilities to generated response token.

WHY IT MATTERS
This definition lowers the mystery. The model can be powerful without being a mind, database, or hand-written rulebook.

BRAIN METAPHOR
Like a person using context to anticipate what word might come next.

VISUAL AID
llm-overview

FEEDBACK
An LLM uses learned weights to turn context into next-token probabilities. Choice notes: A conscious reader: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook. A hand-written rulebook: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook. A search engine that knows every file: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.



- Prompt to Prediction**
Prompt/context flows through learned weights into next-token probabilities; one generated response token is appended.
- 1 Point 1 Prompt/context is given input.
 - 2 Point 2 Learned weights are used during inference.
 - 3 Point 3 The selected response token is appended.

2 / 28

Rationalists vs Empiricists

Rules versus learned patterns

BEFORE MORNING

Stage: architecture Priority: medium Rubric avg: 3.6/5

DEFINITION

Symbolic AI tries to reason with explicit rules and symbols; deep learning learns useful patterns from data by adjusting weights.

DURABLE VS TEMPORARY

Deep learning changes weights during training. Symbolic rules are usually written or configured explicitly.

RELATIONSHIP

If LLMs are learned systems, then the next question is: how are they learned? That leads to training.

BRAIN LIMIT

An LLM is not a full human mind combining logic, experience, emotion, and embodiment. It is a learned prediction system.

CURRENT EXERCISE

durable—or-temporary: Durable or Temporary?

REWRITE NOTES

Risk: LLMs are hand-coded rulebooks, or symbolic AI is obsolete. **Missing:** Early AI often followed the rationalist dream: intelligence as explicit rules, logic, symbols, and search. Deep learning follows a more empiricist path: expose a model to many examples, measure error, and let the system learn internal representations that work. LLMs mostly belong to the deep-learning tradition, although real systems often combine learned models with rules, tools, retrieval, filters, and policies.

CORE EXPLANATION

Early AI often followed the rationalist dream: intelligence as explicit rules, logic, symbols, and search. Deep learning follows a more empiricist path: expose a model to many examples, measure error, and let the system learn internal representations that work. LLMs mostly belong to the deep-learning tradition, although real systems often combine learned models with rules, tools, retrieval, filters, and policies.

PROMPT VS RESPONSE

A prompt is not matched against one giant rulebook. It is processed through learned parameters that shape response-token probabilities.

METAPHOR

A rulebook versus a trained landscape.

MISCONCEPTION

LLMs are hand-coded rulebooks, or symbolic AI is obsolete.

CHECKPOINT

Which phrase best describes the LLM tradition? Correct: Learned patterns from data. Incorrect: Only fixed if-then rules; A spreadsheet macro; A conscious reasoning engine.

ILLUSTRATION NEEDED

Current visual aid: traditions. Split-panel rulebook versus learned landscape, with a small note that systems can combine them.

WHERE IT HAPPENS

This is historical and architectural background. It explains what kind of system an LLM is.

WHY IT MATTERS

This helps learners stop expecting LLMs to behave like ordinary hand-written programs.

BRAIN METAPHOR

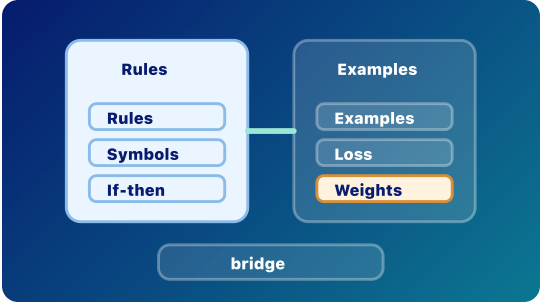
Humans use both explicit reasoning and learned intuition.

VISUAL AID

traditions

FEEDBACK

LLMs are learned prediction systems shaped by data. Choice notes: Only fixed if-then rules: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights. A spreadsheet macro: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights. A conscious reasoning engine: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.



Rules and Learned Patterns

Two traditions, one modern toolkit

Symbolic/rationalist systems use explicit rules; deep-learning/empiricist systems learn useful patterns from examples.

- 1 **Rules** Symbolic systems use explicit if/then logic and symbols.
- 2 **Examples** Deep-learning systems use data, loss, and weight updates.
- 3 **Bridge** Modern AI products can combine learned models with rules and tools.

Key takeaway: Modern AI often blends learned patterns with hand-built rules and tools.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 3/5	Exercise 2/5	Mobile 4/5

rationalism empiricism symbolic-ai deep-learning training weight

Training

Prediction error updates weights

Stage: pretraining

Priority: low

Rubric avg: 4.3/5

DEFINITION

Training is the process that adjusts model weights by comparing predictions with target answers, such as actual next tokens.

DURABLE VS TEMPORARY

Training durably changes weights. Ordinary inference does not.

RELATIONSHIP

Pretraining is the broad first version of this training loop.

BRAIN LIMIT

The model is not reflecting on mistakes. An optimizer changes numbers to reduce error.

CURRENT EXERCISE

training-nudge: Training Nudge

REWRITE NOTES

Risk: When I chat with the model, it automatically trains itself. **Missing:** During training, the model sees examples, predicts something, measures how wrong it was, and updates its weights to do slightly better next time. This happens over many examples and many updates. Training is where durable model capability is created.

CORE EXPLANATION

During training, the model sees examples, predicts something, measures how wrong it was, and updates its weights to do slightly better next time. This happens over many examples and many updates. Training is where durable model capability is created.

PROMPT VS RESPONSE

Training examples may look like text prompts and completions, but the purpose is different: the system is updating weights, not merely producing a live response for a user.

METAPHOR

Practice that changes the instrument before the performance.

MISCONCEPTION

When I chat with the model, it automatically trains itself.

CHECKPOINT

What makes training different from inference? Correct: Training durably updates weights. Incorrect: Training only reads the current context window; Training is the same as sampling; Training writes the final answer.

ILLUSTRATION NEEDED

Current visual aid: training-loop. Training loop with predict, compare, loss, update weights, repeat.

WHERE IT HAPPENS

Before ordinary use, in a separate optimization process.

WHY IT MATTERS

This is the dividing line between learning and using. Many AI myths come from confusing training with inference.

BRAIN METAPHOR

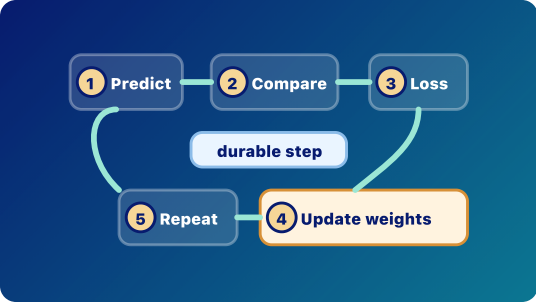
Like practice shaping future habits or skill.

VISUAL AID

training-loop

FEEDBACK

Training changes the model; inference uses the model. Choice notes: Training only reads the current context window: Not quite. Training is a weight-changing process, not ordinary response generation. Training is the same as sampling: Not quite. Training is a weight-changing process, not ordinary response generation. Training writes the final answer: Not quite. Training is a weight-changing process, not ordinary response generation.



Training Loop

Durable change happens at weight update

Predict, compare, loss, update weights, repeat. Training changes weights.

- 1

Predict The model predicts a target.
- 2

Compare The prediction is compared with the target.
- 3

Loss Loss measures error.
- 4

Update weights Weight updates are the durable-change step.
- 5

Repeat The loop repeats many times.

Key takeaway:
Training changes weights; ordinary inference does not.

Rubric Scores				
Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

training

loss

weight

weight-update

parameter

training-data

inference

forward pass

Pretraining

Broad learning before normal use

Stage: pretraining

Priority: low

Rubric avg: 4.1/5

DEFINITION

Pretraining is the broad first training phase where a model learns general patterns by repeatedly predicting tokens across large datasets.

DURABLE VS TEMPORARY

Pretraining durably changes weights.

RELATIONSHIP

Pretraining builds broad capability. Overfitting explains a failure mode: learning examples too narrowly instead of patterns that generalize.

BRAIN LIMIT

Humans connect learning to lived experience. A model stores statistical structure in parameters.

CURRENT EXERCISE

training-nudge: Training Nudge

REWRITE NOTES

Risk: Pretraining means the model can perfectly recall everything it saw. **Missing:** Before a model can be useful in ordinary conversation, it is pretrained on large collections of text and other data. The model repeatedly predicts tokens, measures error, and updates weights. Over time it learns grammar, styles, facts, associations, reasoning patterns, and task shapes. This does not mean it stores every source as a perfect memory.

CORE EXPLANATION

Before a model can be useful in ordinary conversation, it is pretrained on large collections of text and other data. The model repeatedly predicts tokens, measures error, and updates weights. Over time it learns grammar, styles, facts, associations, reasoning patterns, and task shapes. This does not mean it stores every source as a perfect memory.

PROMPT VS RESPONSE

Pretraining creates the weights later used to process prompts and generate responses.

METAPHOR

Billions of tiny raindrops carving paths through a landscape.

MISCONCEPTION

Pretraining means the model can perfectly recall everything it saw.

CHECKPOINT

During pretraining, what changes? Correct: Model weights. Incorrect: Only the current chat; Only the user interface; Nothing changes.

ILLUSTRATION NEEDED

Current visual aid: pretraining-rain. Broad data rain carving durable pathways into a model landscape.

WHERE IT HAPPENS

Before deployment, during the broad foundation-building stage.

WHY IT MATTERS

Pretraining is why next-token prediction can become surprisingly capable. It installs broad pattern knowledge into weights.

BRAIN METAPHOR

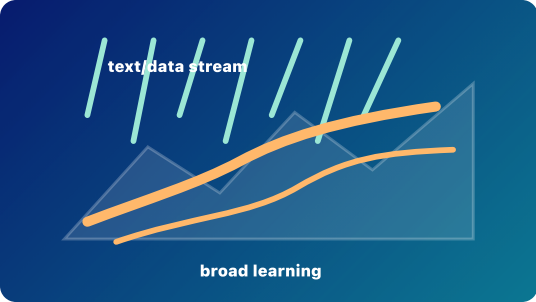
Like broad education shaping later intuition.

VISUAL AID

pretraining-rain

FEEDBACK

Pretraining changes weights before normal use. Choice notes: Only the current chat: Not quite. Pretraining is a durable training process, not a temporary chat event. Only the user interface: Not quite. Pretraining is a durable training process, not a temporary chat event. Nothing changes: Not quite. Pretraining is a durable training process, not a temporary chat event.



Broad Pretraining

A wide stream of data shapes broad durable learning before normal use.

- 1 Point 1 Many examples flow through the training loop.
- 2 Point 2 Repeated updates shape broad model capability.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

pretraining

training

training-data

weight

loss

next-token

inference

Overfitting and Generalization

Memorizing examples versus learning patterns

Stage: pretraining Priority: high Rubric avg: 4.4/5

DEFINITION

Overfitting happens when a model learns training examples too narrowly and performs poorly on new examples; generalization means learned patterns work beyond the examples seen during training.

DURABLE VS TEMPORARY

Overfitting is about what gets installed into weights during training. It is not a temporary inference effect.

RELATIONSHIP

Pretraining should build broad patterns. Fine-tuning must adapt behavior without making the model too narrow.

BRAIN LIMIT

The model is not consciously memorizing. Optimization can make it rely on brittle numerical patterns.

CURRENT EXERCISE

durable—or-temporary: Durable or Temporary?

REWRITE NOTES

Risk: If a model performs well on training data, it has learned well. **Missing:** A model can fit old examples very well but still fail on new cases. That is overfitting. Good learning captures patterns that transfer. For LLMs, generalization is why a model can answer prompts it has never seen exactly before. Overfitting is why evaluation needs unseen examples, not just training performance.

CORE EXPLANATION

A model can fit old examples very well but still fail on new cases. That is overfitting. Good learning captures patterns that transfer. For LLMs, generalization is why a model can answer prompts it has never seen exactly before. Overfitting is why evaluation needs unseen examples, not just training performance.

PROMPT VS RESPONSE

A model that generalizes well can handle new prompts. A model that overfits may repeat brittle patterns from training.

METAPHOR

Studying only the answer key versus learning the subject.

MISCONCEPTION

If a model performs well on training data, it has learned well.

CHECKPOINT

Which is a sign of overfitting? Correct: Great performance on training examples but poor performance on new examples. Incorrect: A model using a context window; A model retrieving documents with RAG; A model generating one token at a time.

ILLUSTRATION NEEDED

Current visual aid: overfitting-generalization. Two curves: an overfit curve that traces training dots too tightly and a smoother curve that predicts new dots.

WHERE IT HAPPENS

During training, evaluation, and model selection.

WHY IT MATTERS

This helps learners see why more memorization is not the same as better intelligence.

BRAIN METAPHOR

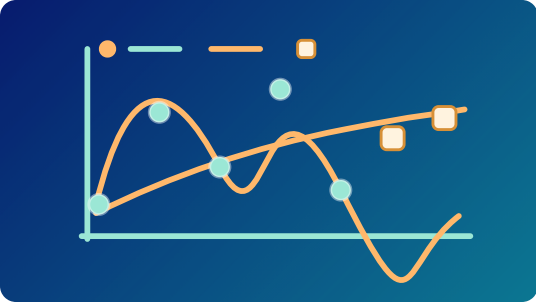
Like memorizing practice questions without understanding the principle.

VISUAL AID

overfitting-generalization

FEEDBACK

Generalization matters more than memorizing the answer key. Choice notes: A model using a context window: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval. A model retrieving documents with RAG: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval. A model generating one token at a time: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.



Overfitting vs Generalization

Memorizing examples is not the same as learning transferable patterns.

- 1 Point 1 Overfit curve traces old dots too tightly.
- 2 Point 2 Generalizing curve is smoother and reaches new examples.
- 3 Point 3 Held-out examples test whether learning transfers.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 5/5	Exercise 4/5	Mobile 4/5

overfitting generalization validation-data training-data evaluation training

Targeted shaping after pretraining

Stage: fine-tuning Priority: low Rubric avg: 4.1/5

DEFINITION

Fine-tuning is additional training that adapts a pretrained model toward a task, domain, style, or preference.

DURABLE VS TEMPORARY

Fine-tuning can durably change weights or add durable adapter behavior. It is different from a prompt, which usually only steers the current run.

RELATIONSHIP

Fine-tuning adapts a model; alignment is one major reason to fine-tune or otherwise shape behavior.

BRAIN LIMIT

The model is not choosing a new identity or values. Training changes output patterns.

CURRENT EXERCISE

durable-or-temporary: Durable or Temporary?

REWRITE NOTES

Risk: Prompting once is the same as fine-tuning.
Missing: A pretrained model knows broad language patterns, but it may not behave the way users need. Fine-tuning adds targeted examples or preference data so future responses are more helpful, specialized, safer, or better aligned with a task. Fine-tuning can update model weights directly or add smaller adapter weights.

CORE EXPLANATION

A pretrained model knows broad language patterns, but it may not behave the way users need. Fine-tuning adds targeted examples or preference data so future responses are more helpful, specialized, safer, or better aligned with a task. Fine-tuning can update model weights directly or add smaller adapter weights.

PROMPT VS RESPONSE

Fine-tuning shapes future response patterns. It does not generate the current response token by token.

METAPHOR

Carving useful trails through an already huge terrain.

MISCONCEPTION

Prompting once is the same as fine-tuning.

CHECKPOINT

Which action is closest to fine-tuning? Correct: Additional training that shapes future responses. Incorrect: Typing one better prompt; Retrieving one PDF into context; Sampling the next token.

ILLUSTRATION NEEDED

Current visual aid: fine-tune-path. Targeted trail added onto an existing pretrained terrain.

WHERE IT HAPPENS

After pretraining and before or between deployment cycles.

WHY IT MATTERS

This explains how one base model can become many differently behaving assistants.

BRAIN METAPHOR

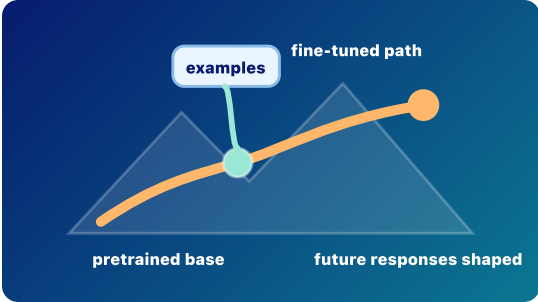
Like coaching someone to follow a house style or professional norm.

VISUAL AID

fine-tune-path

FEEDBACK

Fine-tuning changes future behavior more durably than a single prompt. Choice notes: Typing one better prompt: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training. Retrieving one PDF into context: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training. Sampling the next token: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.



Fine-Tuning Path
Targeted examples nudge an already-trained model toward a desired pattern.

- 1 Point 1 Fine-tuning starts from a pretrained base.
- 2 Point 2 Targeted examples shape future responses.

Rubric Scores				
Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5
fine-tuning adapter pretraining training prompt response alignment				

Alignment

Shaping behavior toward human goals

Stage: alignment

Priority: high

Rubric avg: 4.5/5

DEFINITION

Alignment is the effort to shape AI behavior toward human instructions, preferences, safety boundaries, and values.

DURABLE VS TEMPORARY

Some alignment work durably changes weights or adapter weights. Some alignment happens through system prompts, filters, retrieval, tool rules, or runtime safeguards that steer behavior temporarily.

RELATIONSHIP

Fine-tuning can shape model behavior. Alignment asks: toward what goals and constraints should that behavior be shaped?

BRAIN LIMIT

The model does not acquire human values, moral agency, conscience, or care. Its behavior is shaped by training and system constraints.

CURRENT EXERCISE

durable-or-temporary: Durable or Temporary?

CORE EXPLANATION

A model trained only to predict likely text may produce outputs that are fluent but unhelpful, unsafe, biased, or misleading. Alignment uses methods such as instruction tuning, human feedback, preference optimization, policies, filters, evaluations, and system design to shape model behavior. Alignment is not one switch. It is an ongoing effort to make model behavior more useful, reliable, and compatible with human goals.

PROMPT VS RESPONSE

Alignment affects how the model responds to prompts, but it does not mean the model understands morality like a person.

METAPHOR

Guardrails, coaching, and a compass.

MISCONCEPTION

Aligned means morally good or fully trustworthy.

CHECKPOINT

What does alignment try to do? Correct: Shape model behavior toward human goals, instructions, and safety boundaries. Incorrect: Make the model conscious; Guarantee every answer is true; Permanently solve all AI risk.

WHERE IT HAPPENS

During fine-tuning, human feedback learning, system design, evaluation, deployment, and policy enforcement.

WHY IT MATTERS

Alignment is why a model may refuse harmful requests, follow instructions, or prefer clearer helpful answers over merely likely text.

BRAIN METAPHOR

Like learning social norms or professional expectations.

VISUAL AID

alignment

FEEDBACK

Alignment shapes behavior, but it is not magic morality. Choice notes: Make the model conscious: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee. Guarantee every answer is true: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee. Permanently solve all AI risk: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.

Alignment Landscape

Alignment shapes behavior with preferred paths, guardrails, feedback, and policies; it does not create moral agency.

- Point 1** Preferred path: behavior the system encourages.
- Point 2** Guardrail: boundary around risky terrain.
- Point 3** Warning zone: outputs to avoid or handle carefully.
- Point 4** Policy check and feedback are system signals, not conscience.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 5/5	Visual 5/5	Exercise 4/5	Mobile 4/5

- alignment
- instruction-tuning
- human-feedback-learning
- rlhf
- preference-optimization
- policy
- guardrail
- evaluation
- fine-tuning

Inference

A forward pass, not a memory update

Stage: inference

Priority: low

Rubric avg: 4.4/5

DEFINITION

Inference is normal model use: fixed learned weights process the current context to produce scores for the next response token.

DURABLE VS TEMPORARY

Inference creates temporary hidden states and next-token scores for this run. It normally does not durably change the model weights.

RELATIONSHIP

Batch 1 explained how weights are shaped before use. Inference is the live run that uses those weights without rewriting them.

BRAIN LIMIT

The model is not conscious or living; it computes temporary vectors with fixed weights.

CURRENT EXERCISE

training–nudge: Training Nudge

REWRITE NOTES

Risk: Normal chat automatically trains the model.
Missing: During ordinary use, the current context enters the model, embeddings and hidden states are computed, the final hidden state produces next-token scores, and decoding chooses one response token. Then the token is appended and the process repeats. The learned weights are used, but they normally stay fixed.

CORE EXPLANATION

During ordinary use, the current context enters the model, embeddings and hidden states are computed, the final hidden state produces next-token scores, and decoding chooses one response token. Then the token is appended and the process repeats. The learned weights are used, but they normally stay fixed.

PROMPT VS RESPONSE

Prompt tokens, retrieved context, conversation history, and response-so-far tokens can all be current context; inference generates the next response token.

METAPHOR

Using a map to choose a route, not redrawing the map.

MISCONCEPTION

Normal chat automatically trains the model.

CHECKPOINT

What changes during ordinary inference? Correct: Temporary hidden states. Incorrect: The model weights permanently; The training dataset; Every document the model has seen.

ILLUSTRATION NEEDED

Current visual aid: inference-pass. Forward-pass pipeline with fixed weights underneath and temporary states above.

WHERE IT HAPPENS

Every time a deployed model answers a prompt.

WHY IT MATTERS

This is the line between using a trained model and training a model.

BRAIN METAPHOR

Like using working memory to answer from the information currently in front of you.

VISUAL AID

inference–pass

FEEDBACK

Inference uses fixed weights and creates temporary states for this run. Choice notes: The model weights permanently: Not quite. Ordinary inference normally does not rewrite weights or training data. The training dataset: Not quite. Ordinary inference normally does not rewrite weights or training data. Every document the model has seen: Not quite. Ordinary inference normally does not rewrite weights or training data.

1

context

2

weights fixed

3

states

scores

next

Forward Pass
Inference creates temporary states while durable weights stay fixed.

1

Point 1

Current context enters the model.

2

Point 2

Fixed weights are used, not updated.

3

Point 3

Temporary hidden states lead to next-token scores.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

inference

forward-pass

input-context

weight

hidden state

logits

sampling

Prompt vs Response

Given context versus generated tokens

Stage: response-generation

Priority: low

Rubric avg: 4.4/5

DEFINITION

Prompt tokens are given to the model; response tokens are generated by the model one at a time and appended to the context.

CORE EXPLANATION

The user prompt should be a complete request, such as: "Give me one sentence with two household pets in conflict." The model then begins generating a response. During generation, the response-so-far may be incomplete: "A jealous dog chased a startled cat across the kitchen". After the model chooses one response token, such as "floor", that token is appended. The next forward pass sees the user prompt plus the response so far.

WHERE IT HAPPENS

During inference and response generation.

DURABLE VS TEMPORARY

Prompt and response-so-far tokens are temporary context. They can influence the current run, but they do not normally update weights.

PROMPT VS RESPONSE

This is the distinction itself: the user prompt is given context; "floor" is a generated response token appended to the response-so-far.

WHY IT MATTERS

It prevents the myth that the model writes a full answer all at once.

RELATIONSHIP

After we distinguish prompt from response, we can see how both are represented as tokens.

METAPHOR

Given cards on a table, then new cards added one at a time.

BRAIN METAPHOR

Like hearing a sentence and continuing it word by word.

BRAIN LIMIT

A person may plan, intend, and revise. The model repeatedly samples next tokens from probabilities.

MISCONCEPTION

The model generates the whole response at once.

VISUAL AID

prompt-response

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

CHECKPOINT

What happens after one response token is generated? Correct: It is appended to the context. Incorrect: The whole answer appears at once; The model permanently learns it; Softmax disappears.

FEEDBACK

The response grows by next token, append, repeat. Choice notes: The whole answer appears at once: Not quite. A generated token becomes part of the next context, but it is not a durable weight update. The model permanently learns it: Not quite. A generated token becomes part of the next context, but it is not a durable weight update. Softmax disappears: Not quite. A generated token becomes part of the next context, but it is not a durable weight update.

REWRITE NOTES

Risk: The model generates the whole response at once. **Missing:** The user prompt should be a complete request, such as: "Give me one sentence with two household pets in conflict." The model then begins generating a response. During generation, the response-so-far may be incomplete: "A jealous dog chased a startled cat across the kitchen". After the model chooses one response token, such as "floor", that token is appended. The next forward pass sees the user prompt plus the response so far.

ILLUSTRATION NEEDED

Current visual aid: prompt-response. Given prompt cards and newly generated response cards entering the next context together.



Prompt vs Response
Given context versus generated tokens

A complete user prompt is given; response tokens are generated and appended.

1

User prompt

The complete request is: Give me one sentence with two household pets in conflict.

2

Response so far

The model has already generated: A jealous dog chased a startled cat across the kitchen.

3

Next token

floor is appended, so the next context combines prompt plus response so far plus floor.

Key takeaway:

The response grows one token at a time inside the current context.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

prompt

response

prompt-tokens

response-tokens

generated-token

input-context

completion

Text becomes model-readable chunks

Stage: prompt-processing

Priority: medium

Rubric avg: 4.2/5

DEFINITION

Tokenization splits text into chunks that can be represented as token IDs.

DURABLE VS TEMPORARY

Tokenization itself does not change model weights. It is a representation step during input/output processing.

RELATIONSHIP

Tokens become token IDs, and token IDs retrieve embeddings.

BRAIN LIMIT

Human language perception uses sound, meaning, memory, and experience. Tokenization uses tokenizer rules and token IDs.

CURRENT EXERCISE

prompt-or-response-label: Prompt or Response?

REWRITE NOTES

Risk: Tokens are always whole words or human concepts. **Missing:** LLMs do not carry raw English sentences through their layers. Text is first split into tokens: pieces that might be whole words, word parts, punctuation, or spaces. The generated response "A jealous dog chased a startled cat across the kitchen floor." can be shown as simplified token cards for learning. Real tokenizers may split it differently. Prompt text is also tokenized before model processing, and generated response tokens are token IDs before they are displayed as text.

CORE EXPLANATION

LLMs do not carry raw English sentences through their layers. Text is first split into tokens: pieces that might be whole words, word parts, punctuation, or spaces. The generated response "A jealous dog chased a startled cat across the kitchen floor." can be shown as simplified token cards for learning. Real tokenizers may split it differently. Prompt text is also tokenized before model processing, and generated response tokens are token IDs before they are displayed as text.

PROMPT VS RESPONSE

Both prompt text and generated response text are represented as tokens. Generated response tokens are chosen as token IDs, then displayed as text.

METAPHOR

Parcels on a conveyor belt entering the model.

MISCONCEPTION

Tokens are always whole words or human concepts.

CHECKPOINT

What can a token be? Correct: A word, word piece, punctuation mark, or other text chunk. Incorrect: Only a complete word; A permanent memory; A whole finished answer.

ILLUSTRATION NEEDED

Current visual aid: tokenization. Tokenizer conveyor splitting a short sentence into uneven chunks.

WHERE IT HAPPENS

Before embedding lookup and before transformer layers process the current context.

WHY IT MATTERS

Without tokenization, the model has no numerical pieces to process.

BRAIN METAPHOR

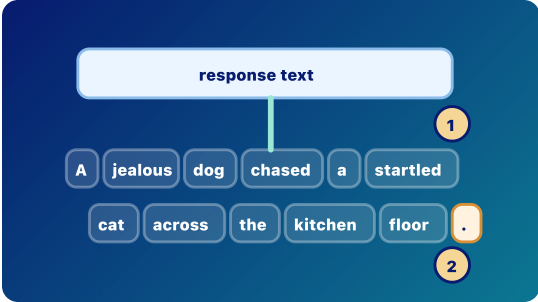
Like hearing a sentence in meaningful chunks before interpreting it.

VISUAL AID

tokenization

FEEDBACK

Tokens are model-readable chunks, not always words. Choice notes: Only a complete word: Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries. A permanent memory: Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries. A whole finished answer: Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.



Text to Tokens

Generated response text is split into model-readable chunks before embedding lookup.

1 **Point 1** Generated response: A jealous dog chased a startled cat across the kitchen floor.

2 **Point 2** Simplified for learning. Real tokenizers may split differently.

Rubric Scores				
Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

tokenization

tokenizer

token

prompt-tokens

response-tokens

token-id

embedding

Token IDs

Chunks become lookup numbers

Stage: prompt-processing Priority: low Rubric avg: 4.1/5

DEFINITION
A token ID is the number assigned to a token so the model can look up its learned starting vector.

DURABLE VS TEMPORARY
Token IDs are part of representation during inference. The mapping is fixed by the tokenizer/model setup; using an ID does not update weights.

RELATIONSHIP
Token IDs point to rows in the embedding table.

BRAIN LIMIT
Humans do not assign fixed integer IDs to every language chunk. The number itself is not understanding.

CURRENT EXERCISE
prompt-or-response-label: Prompt or Response?

REWRITE NOTES
Risk: The token ID is the meaning. **Missing:** After tokenization, each token is represented by an integer ID. The ID is not the meaning. It is a lookup key. For this app example, dog maps to 421, cat maps to 982, and floor maps to 1576. The embedding table uses each ID to retrieve a learned starting vector. During generation, the model also chooses a next token ID before it is displayed as text.

CORE EXPLANATION
After tokenization, each token is represented by an integer ID. The ID is not the meaning. It is a lookup key. For this app example, dog maps to 421, cat maps to 982, and floor maps to 1576. The embedding table uses each ID to retrieve a learned starting vector. During generation, the model also chooses a next token ID before it is displayed as text.

PROMPT VS RESPONSE
Prompt tokens become token IDs before processing. Generated response tokens are also token IDs before being shown as text and appended to context.

METAPHOR
A library call number.

MISCONCEPTION
The token ID is the meaning.

CHECKPOINT
Why does the model need token IDs? Correct: To look up embeddings. Incorrect: To store a permanent memory; To skip tensors; To guarantee truth.

ILLUSTRATION NEEDED
Current visual aid: token-ids. Token chunks mapped to numeric IDs, then to rows in an embedding table.

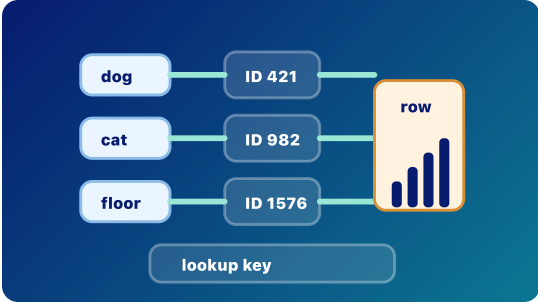
WHERE IT HAPPENS
Between tokenization and embedding lookup.

WHY IT MATTERS
Token IDs are the bridge from text chunks to numerical computation.

BRAIN METAPHOR
A little like recognizing a symbol and activating a familiar association.

VISUAL AID
token-ids

FEEDBACK
Token IDs connect text chunks to learned embedding vectors. Choice notes: To store a permanent memory: Not quite. A token ID is a lookup key, not meaning, memory, or truth. To skip tensors: Not quite. A token ID is a lookup key, not meaning, memory, or truth. To guarantee truth: Not quite. A token ID is a lookup key, not meaning, memory, or truth.



- Token IDs**
Each token gets a lookup number that points to an embedding row.
- 1 **Point 1** dog -> 421, cat -> 982, floor -> 1576.
 - 2 **Point 2** The ID is a lookup key, not the meaning.
 - 3 **Point 3** The ID points to an embedding row.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

token-id token tokenizer embedding-table embedding

Token IDs look up learned starting vectors

Stage: prompt-processing

Priority: low

Rubric avg: 4.3/5

DEFINITION

An embedding is a learned starting vector retrieved for a token ID.

DURABLE VS TEMPORARY

The embedding table is durable learned model data. The retrieved embedding vector is used temporarily during a forward pass.

RELATIONSHIP

Embeddings are vectors. The next lesson explains vectors more generally.

BRAIN LIMIT

A human association includes memory, sensation, and experience. An embedding is a learned row of numbers.

CURRENT EXERCISE

prompt-or-response-label: Prompt or Response?

REWRITE NOTES

Risk: An embedding is a dictionary definition or a human memory. **Missing:** The embedding table is learned during training. At inference time, each token ID in the current context retrieves one row from that table: for example, floor -> 1576 -> one starting vector. That row is the token's starting vector before context changes it. Later layers turn embeddings into hidden states shaped by surrounding tokens.

CORE EXPLANATION

The embedding table is learned during training. At inference time, each token ID in the current context retrieves one row from that table: for example, floor -> 1576 -> one starting vector. That row is the token's starting vector before context changes it. Later layers turn embeddings into hidden states shaped by surrounding tokens.

PROMPT VS RESPONSE

Prompt tokens and response-so-far tokens are looked up as embeddings when they are inside the current context. A newly generated token will get its embedding during the next forward pass after it is appended.

METAPHOR

A token gets a starting address in a giant meaning cloud.

MISCONCEPTION

An embedding is a dictionary definition or a human memory.

CHECKPOINT

What is an embedding in this app? Correct: A learned starting vector for a token ID. Incorrect: A finished sentence; A human memory; A secret English definition.

ILLUSTRATION NEEDED

Current visual aid: embeddings. Lookup table pulling a learned starting vector for one token ID.

WHERE IT HAPPENS

At the start of model processing, before transformer layers.

WHY IT MATTERS

Embeddings are the first step from token IDs into the model's learned numerical space.

BRAIN METAPHOR

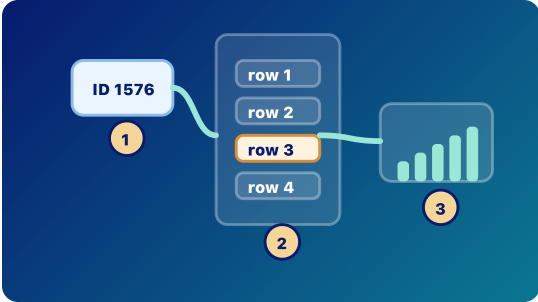
Like the first associations that light up when you hear a word.

VISUAL AID

embeddings

FEEDBACK

An embedding is a learned starting vector before context reshapes it. Choice notes: A finished sentence: Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text. A human memory: Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text. A secret English definition: Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.



Embedding Lookup

A token ID retrieves a learned starting vector.

- Point 1** The embedding table was learned during training.
- Point 2** Inference retrieves one row temporarily for the current context.
- Point 3** The retrieved embedding is a starting vector, not a definition.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

embedding

embedding-table

token-id

vector

hidden state

inference

training

Lists of numbers carrying features

Stage: architecture

Priority: high

Rubric avg: 3.8/5

DEFINITION

A vector is a list of numbers that represents features of something, such as a token.

DURABLE VS TEMPORARY

Some vectors are durable learned parameters, such as embedding rows. Many vectors during inference are temporary activations, such as hidden states.

RELATIONSHIP

Several token vectors organized together become a tensor.

BRAIN LIMIT

The model's vector dimensions are learned numerical patterns, not conscious associations or human-labeled ideas.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Each vector dimension has a simple human-readable label. **Missing:** In an LLM, many things are represented as vectors. An embedding is a starting vector for a token ID. A hidden state is a later vector shaped by context. The numbers are not usually neat labeled features like animal-ish, grammar role, tone, position, or context clue. Those labels are teaching simplifications. Real features are distributed across many dimensions, which is why vectors can represent fuzzy relationships.

CORE EXPLANATION

In an LLM, many things are represented as vectors. An embedding is a starting vector for a token ID. A hidden state is a later vector shaped by context. The numbers are not usually neat labeled features like animal-ish, grammar role, tone, position, or context clue. Those labels are teaching simplifications. Real features are distributed across many dimensions, which is why vectors can represent fuzzy relationships.

PROMPT VS RESPONSE

Prompt tokens and response-so-far tokens become vectors when processed in the current context. Those vectors help choose the next response token, such as "floor".

METAPHOR

A dashboard of many sliders.

MISCONCEPTION

Each vector dimension has a simple human-readable label.

CHECKPOINT

Which statement is most accurate? Correct: Embeddings and hidden states are both vectors. Incorrect: Vectors are whole paragraphs; Vectors are permanent chats; Every vector dimension is a neat English meaning.

ILLUSTRATION NEEDED

Current visual aid: vectors. Many-dimensional feature bars with a warning that labels are simplified.

WHERE IT HAPPENS

Throughout the model: embeddings, hidden states, attention calculations, MLP transformations, and output scoring all involve vectors.

WHY IT MATTERS

Vectors let the model compute with many fuzzy features at once instead of using plain-English labels.

BRAIN METAPHOR

Like many weak associations around a word lighting up together.

VISUAL AID

vectors

FEEDBACK

Vectors are lists of numbers used at many points in the model. Choice notes: Vectors are whole paragraphs: Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled. Vectors are permanent chats: Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled. Every vector dimension is a neat English meaning: Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.

simplified

1

animal-ish

grammar role

tone

position

context clue

2

distributed features

Feature Vector

A vector is a list of numerical features, not a sentence.

1

Point 1

Labels such as animal-ish, grammar role, tone, position, and context clue are simplified.

2

Point 2

Real features are distributed across many dimensions.

Rubric Scores				
Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 3/5	Exercise 2/5	Mobile 4/5

vector

feature

distributed-representation

embedding

hidden state

activation

Tensors

Organized blocks of vectors

Stage: architecture

Priority: high

Rubric avg: 3.7/5

DEFINITION

A tensor is an organized block of numbers, often holding many token vectors at once.

DURABLE VS TEMPORARY

Weight tensors are durable learned parameters. Activation tensors during inference are temporary.

RELATIONSHIP

Now that current context is in tensor form, transformer layers can apply attention and MLPs. That leads into Workday.

BRAIN LIMIT

The brain is not literally storing rectangular arrays with token and feature axes.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: A tensor is one word, one number, or a mysterious object. **Missing:** Once tokens have embeddings, the model needs to carry many token positions and many features together. A simple way to picture this is a table: token positions down one side, feature numbers across the other. With batches, heads, and layers, the shapes get richer. But the core idea is simple: tensors organize vectors so transformer layers can process the current context.

CORE EXPLANATION

Once tokens have embeddings, the model needs to carry many token positions and many features together. A simple way to picture this is a table: token positions down one side, feature numbers across the other. With batches, heads, and layers, the shapes get richer. But the core idea is simple: tensors organize vectors so transformer layers can process the current context.

PROMPT VS RESPONSE

The current context tensor may include the complete user prompt, retrieved tokens, prior conversation tokens, and response-so-far tokens such as "A jealous dog chased a startled cat across the kitchen".

METAPHOR

A stack of spreadsheets carrying token features through the model.

MISCONCEPTION

A tensor is one word, one number, or a mysterious object.

CHECKPOINT

Which is the better definition of tensor here? Correct: An organized block of numbers. Incorrect: A word in a dictionary; A rule written by a programmer; A permanent memory of the prompt.

ILLUSTRATION NEEDED

Current visual aid: tensors. Mobile-readable tensor block with token-position and feature axes.

WHERE IT HAPPENS

Tensors carry data through the transformer layers during a forward pass. Model weights are also stored as tensors.

WHY IT MATTERS

The transformer transforms tensors, not raw words.

BRAIN METAPHOR

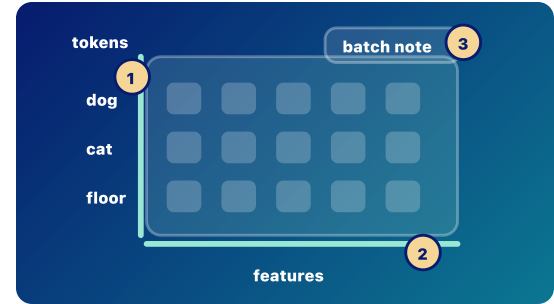
Like keeping many signals active at once.

VISUAL AID

tensors

FEEDBACK

Tensors organize many vectors so the model can compute with them. Choice notes: A word in a dictionary: Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory. A rule written by a programmer: Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory. A permanent memory of the prompt: Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.



Tensor Block
Tensors organize many token vectors for layer-by-layer processing.

- 1 Point 1** Token positions run down one axis.
- 2 Point 2** Feature dimensions run across another axis.
- 3 Point 3** Batch x tokens x features is an advanced shape note.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

tensor

vector

activation

weight-tensor

input-context

layer

Weighted relevance between positions

Stage: architecture

Priority: high

Rubric avg: 4.4/5

DEFINITION

Attention assigns weighted relevance between token positions.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

During inference, attention operates over token positions in the current context, including prompt tokens and earlier generated response tokens.

BRAIN LIMIT

It is not human attention, awareness, desire, or focus. It is weighted relevance in a computation.

CURRENT EXERCISE

attention–relevance: Attention Is Relevance

REWRITE NOTES

Risk: The term attention invites human-focus and awareness metaphors. **Missing:** Show attention weights as arrows between positions, not as a mind selecting ideas.

CORE EXPLANATION

Attention assigns weighted relevance between token positions.

PROMPT VS RESPONSE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

METAPHOR

Spotlights showing which earlier tokens are useful right now.

MISCONCEPTION

The term attention invites human-focus and awareness metaphors.

CHECKPOINT

Which tokens can attention look across during inference? Correct: Tokens currently in the context window. Incorrect: All text the model has ever seen; Only the newest generated token.

ILLUSTRATION NEEDED

Current visual aid: attention. Token nodes with weighted relevance arcs and a plain-language caption.

WHERE IT HAPPENS

It happens inside transformer layers during a forward pass.

WHY IT MATTERS

Attention lets one token position use information from other visible token positions in the current context.

BRAIN METAPHOR

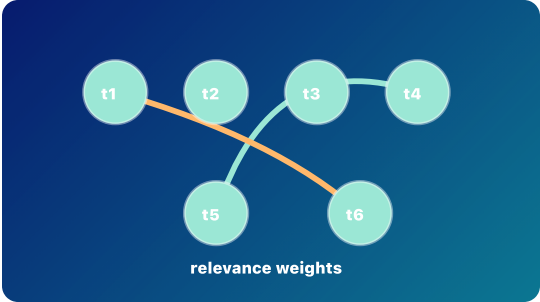
The word attention is useful because some parts of the context matter more than others.

VISUAL AID

attention

FEEDBACK

Attention operates over the current context. It does not browse all training data.



Attention Weave
Attention is weighted relevance between token positions.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 5/5	Visual 4/5	Exercise 4/5	Mobile 4/5

attention

input context

prompt tokens

response tokens

hidden state

layer

Feature reshaping per token

Stage: architecture

Priority: high

Rubric avg: 4.3/5

DEFINITION

An MLP is a feed-forward network that reshapes each token position's feature vector.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

During a forward pass over the current context, attention mixes information across positions and the MLP reshapes features within each position.

BRAIN LIMIT

The MLP is not a neuron with feelings or beliefs; it is a learned function applied to vectors.

CURRENT EXERCISE

mlp-feature-reshape: MLP Feature Reshape

REWRITE NOTES

Risk: The MLP can blur into attention unless cross-position mixing and per-position reshaping are contrasted. **Missing:** Put MLP beside attention in the same layer so their complementary roles are visible.

CORE EXPLANATION

An MLP is a feed-forward network that reshapes each token position's feature vector.

PROMPT VS RESPONSE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

METAPHOR

A forge that bends each token vector into a more useful shape.

MISCONCEPTION

The MLP can blur into attention unless cross-position mixing and per-position reshaping are contrasted.

CHECKPOINT

What does the MLP mainly do here? Correct: Reshapes per-token features. Incorrect: Reads private files; Chooses the final answer directly.

ILLUSTRATION NEEDED

Current visual aid: mlp. Per-token vector forge or feature mixer after attention.

WHERE IT HAPPENS

It appears inside transformer layers, usually after attention has mixed information across positions.

WHY IT MATTERS

The MLP helps turn mixed context into more useful per-token features for later layers.

BRAIN METAPHOR

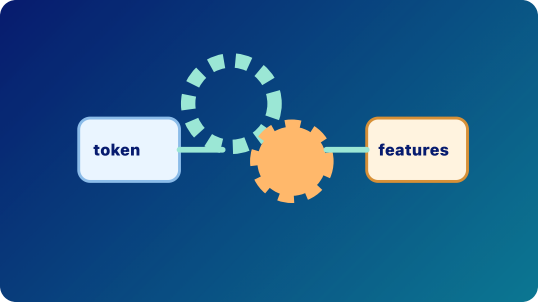
It is somewhat like a small processing step after noticing relevant context.

VISUAL AID

mlp

FEEDBACK

MLPs transform token feature vectors after attention has mixed contextual information.



MLP Reshape
The MLP reshapes each token position feature vector.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 4/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

MLPinput contexthidden stateattention

Stage: architecture

Priority: medium

Rubric avg: 3.9/5

DEFINITION

A transformer layer is a repeated block that usually combines attention, MLP feature reshaping, residual paths, and normalization.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

Each layer refines hidden states while residual paths help useful information carry forward.

BRAIN LIMIT

The layers do not form a human-like chain of thought; they update numerical states.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Layer stacks can be mistaken for a human chain of thought. **Missing:** Add residual paths and normalization as optional advanced labels or a split target lesson.

CORE EXPLANATION

A transformer layer is a repeated block that usually combines attention, MLP feature reshaping, residual paths, and normalization.

PROMPT VS RESPONSE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

METAPHOR

A series of editing passes where each pass can revise but also preserve the draft.

MISCONCEPTION

Layer stacks can be mistaken for a human chain of thought.

CHECKPOINT

What do layers repeatedly refine? Correct: Hidden states. Incorrect: Permanent user memory; The app interface.

ILLUSTRATION NEEDED

Current visual aid: layers. Transparent layer stack with attention and MLP blocks repeated.

WHERE IT HAPPENS

Layers are stacked inside the model between embeddings and next-token scores.

WHY IT MATTERS

Repeated layers let simple starting vectors become richer, context-shaped hidden states.

BRAIN METAPHOR

The stack can feel like stages of processing.

VISUAL AID

Layers

FEEDBACK

Layers update temporary internal vectors during inference.

layer

layer

layer

layer

carry forward

Layer Stack

Repeated blocks refine hidden states while carrying useful signal forward.

Rubric Scores				
Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

layer

attention

MLP

hidden state

Hidden States

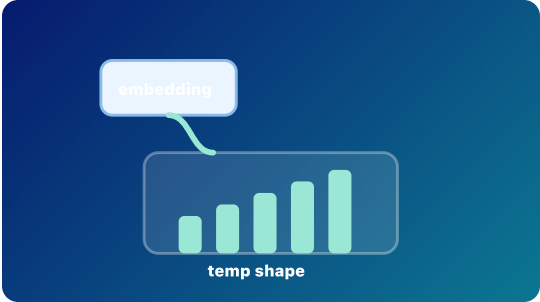
Temporary contextual vectors

Stage: inference

Priority: high

Rubric avg: 4.4/5

DEFINITION A hidden state is the model's temporary internal context-shaped vector for a token at a given layer.	CORE EXPLANATION A hidden state is the model's temporary internal context-shaped vector for a token at a given layer.	WHERE IT HAPPENS Hidden states appear during the forward pass after embeddings enter the layer stack.
DURABLE VS TEMPORARY Not listed in current app data.	PROMPT VS RESPONSE This is the live prompt-to-response path: fixed weights process current context and generate response tokens.	WHY IT MATTERS They are where the prompt's context changes the token representation before the model scores possible next tokens.
RELATIONSHIP Hidden states are created while processing the current context during a forward pass; they are not the visible response.	METAPHOR A scratchpad of numbers that changes while the prompt is being processed.	BRAIN METAPHOR Hidden state can resemble temporary working memory.
BRAIN LIMIT It is not permanent memory, consciousness, or private English text; it is a temporary vector.	MISCONCEPTION Hidden states can be mistaken for secret English text or permanent memory.	VISUAL AID hidden-states
CURRENT EXERCISE mlp-feature-reshape: MLP Feature Reshape	CHECKPOINT Why is it called hidden state? Correct: It is internal numbers, not visible text. Incorrect: It is encrypted English; It is a secret memory file.	FEEDBACK Hidden means inside the model's working representation.
REWRITE NOTES Risk: Hidden states can be mistaken for secret English text or permanent memory. Missing: Show hidden state changing layer by layer for the same token in two contexts.	ILLUSTRATION NEEDED Current visual aid: hidden-states. Same token glowing differently across contexts and layers.	



Temporary Hidden State
Hidden states are context-shaped vectors created during a forward pass.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

hidden state

input context

forward pass

embedding

layer

inference

Raw next-token scores

Stage: response generation

Priority: medium

Rubric avg: 4.4/5

DEFINITION

Logits are raw scores for the next token before they become probabilities.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

A forward pass over the current context produces logits for one next generated token.

BRAIN LIMIT

The model is not weighing options with intent; logits are raw numerical scores.

CURRENT EXERCISE

softmax-funnel: Softmax Funnel

REWRITE NOTES

Risk: Raw scores can be confused with probabilities or truth confidence. **Missing:** Make vocabulary projection explicit: final hidden state to one score per candidate token.

CORE EXPLANATION

Logits are raw scores for the next token before they become probabilities.

PROMPT VS RESPONSE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

METAPHOR

A scoreboard before the points are converted into odds.

MISCONCEPTION

Raw scores can be confused with probabilities or truth confidence.

CHECKPOINT

Where do probabilities come from in this path? Correct: Softmax converts logits. Incorrect: Training rewrites the weights; The model asks a database.

ILLUSTRATION NEEDED

Current visual aid: logits. Raw candidate scoreboard before softmax.

WHERE IT HAPPENS

They appear near the end of a forward pass, after the final hidden state is projected toward the vocabulary.

WHY IT MATTERS

Logits show that the model has not chosen a token yet; it has scored candidates.

BRAIN METAPHOR

It can feel like considering options.

VISUAL AID

logits

FEEDBACK

Logits are raw scores; softmax turns them into probabilities.



Raw Scores
Logits are raw next-token scores before probabilities.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

logits

generated token

forward pass

hidden state

softmax

Softmax

Scores become probabilities

Stage: response generation

Priority: medium

Rubric avg: 4.4/5

DEFINITION

Softmax turns next-token raw scores into probabilities that sum to one.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

Softmax converts logits into probabilities; sampling then chooses one token from those probabilities.

BRAIN LIMIT

The probabilities are mathematical outputs, not confidence, desire, or truth.

CURRENT EXERCISE

softmax-funnel: Softmax Funnel

REWRITE NOTES

Risk: Probabilities can be mistaken for correctness or factual confidence. **Missing:** Show that softmax normalizes scores into a distribution that sums to one.

CORE EXPLANATION

Softmax turns next-token raw scores into probabilities that sum to one.

PROMPT VS RESPONSE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

METAPHOR

A funnel turning scoreboard points into odds.

MISCONCEPTION

Probabilities can be mistaken for correctness or factual confidence.

CHECKPOINT

What does softmax produce? Correct: Next-token probabilities. Incorrect: Permanent memory; A whole essay.

ILLUSTRATION NEEDED

Current visual aid: softmax. Funnel converting raw scores into probabilities that sum to one.

WHERE IT HAPPENS

It sits between logits and sampling during a decoding step.

WHY IT MATTERS

Softmax makes scores comparable as a probability cloud over candidate tokens.

BRAIN METAPHOR

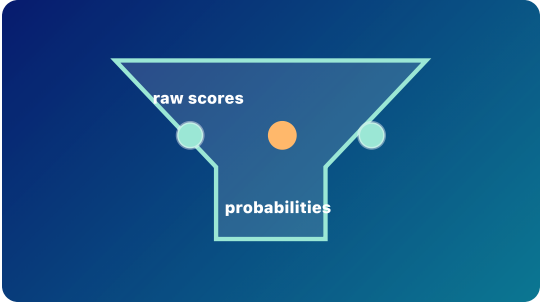
It resembles turning vague preferences into relative chances.

VISUAL AID

softmax

FEEDBACK

Softmax turns raw next-token scores into a probability distribution.



Softmax Funnel
Softmax converts raw scores into probabilities that sum to one.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

softmax

logits

decoding step

sampling

probability

Sampling

One next token is chosen

Stage: response generation

Priority: medium

Rubric avg: 4.3/5

DEFINITION

Sampling chooses one next token for the response from the probability cloud.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

Temperature, top-k, and top-p shape the candidate pool before the chosen response token is appended to the context.

BRAIN LIMIT

The model is not choosing with intention; a decoding rule selects from probabilities.

CURRENT EXERCISE

pick-next-token: Pick the Next Token

REWRITE NOTES

Risk: Randomness can sound arbitrary unless tied to weighted probabilities. **Missing:** Add temperature/top-p as knobs that shape selection without overexplaining math.

CORE EXPLANATION

Sampling chooses one next token for the response from the probability cloud.

PROMPT VS RESPONSE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

METAPHOR

Choosing from a weighted bowl of possible next tokens.

MISCONCEPTION

Randomness can sound arbitrary unless tied to weighted probabilities.

CHECKPOINT

When the model chooses one token, what happens next? Correct: The token is appended and the model runs again. Incorrect: The model writes the whole response at once; The model permanently learns the token.

ILLUSTRATION NEEDED

Current visual aid: sampling. Weighted vocabulary cloud with one selected token.

WHERE IT HAPPENS

It happens after softmax in the decoding step.

WHY IT MATTERS

Sampling explains why the same prompt can sometimes produce different good responses.

BRAIN METAPHOR

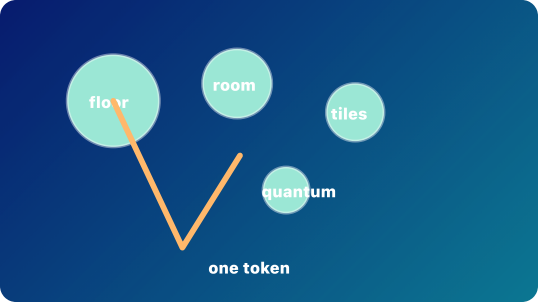
It can resemble choosing a word while speaking.

VISUAL AID

sampling

FEEDBACK

The selected response token becomes part of the context for the next forward pass.



Sample One Token
Sampling chooses one token from the probability cloud.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

sampling

generated token

decoding step

response tokens

temperature

top-k

top-p

Stage: response generation

Priority: low

Rubric avg: 4.4/5

DEFINITION

Autoregression means response generation by repeated next-token prediction: choose one token, append it, and run again.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

Each generated response token becomes part of the input context for the next decoding step.

BRAIN LIMIT

The model is not holding a plan in mind; it repeatedly predicts the next token from visible context.

CURRENT EXERCISE

prompt-or-response-label: Prompt or Response?

REWRITE NOTES

Risk: Learners may still imagine the model drafting a full paragraph internally before showing it. **Missing:** Use the canonical pet-conflict response to show next token, append, repeat.

CORE EXPLANATION

Autoregression means response generation by repeated next-token prediction: choose one token, append it, and run again.

PROMPT VS RESPONSE

This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

METAPHOR

A train adding one car at a time.

MISCONCEPTION

Learners may still imagine the model drafting a full paragraph internally before showing it.

CHECKPOINT

How does an LLM normally generate text? Correct: One token at a time. Incorrect: All words at once; By rewriting its weights.

ILLUSTRATION NEEDED

Current visual aid: autoregression. Loop diagram: score, sample, append, run again.

WHERE IT HAPPENS

It is the loop that builds text during inference.

WHY IT MATTERS

Autoregression explains why earlier generated tokens become part of what the model can use next.

BRAIN METAPHOR

It resembles continuing a thought step by step.

VISUAL AID

autoregression

FEEDBACK

LLM generation is autoregressive: next token, append, repeat.



Append and Repeat
The chosen token is appended, then the model runs again.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

autoregression

completion

generated token

response tokens

sampling

inference

23 / 28

Context Window

Temporary working context

THE DAY REPEATS

Stage: current context

Priority: high

Rubric avg: 4.5/5

DEFINITION

A context window is the limited amount of tokens or media the model can currently use.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

The model can only use tokens currently inside the context window, whether they came from the prompt, prior conversation, retrieved documents, or the response-so-far.

BRAIN LIMIT

It is not durable memory. What falls out cannot be directly attended to in the next pass.

CURRENT EXERCISE

context-window-fell-out: Context Window: What Fell Out?

REWRITE NOTES

Risk: Current context can be confused with saved memory or training data. **Missing:** Show what falls out and that retrieval adds text to context rather than training weights.

CORE EXPLANATION

A context window is the limited amount of tokens or media the model can currently use.

PROMPT VS RESPONSE

This card should separate the temporary visible context from durable memory or weight updates.

METAPHOR

A moving spotlight over a growing train.

MISCONCEPTION

Current context can be confused with saved memory or training data.

CHECKPOINT

What can happen when the context window is full? Correct: Older tokens may be truncated. Incorrect: The model permanently learns them; Softmax disappears.

ILLUSTRATION NEEDED

Current visual aid: context-window. Limited stack of cards where older cards fall out.

WHERE IT HAPPENS

It surrounds the prompt, conversation history, retrieved material, and response-so-far that remain visible.

WHY IT MATTERS

It explains why models can use current context but do not automatically form permanent memory from every chat.

BRAIN METAPHOR

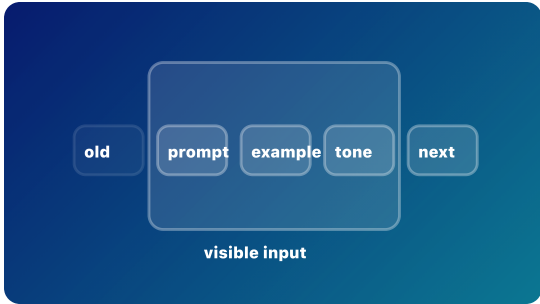
It is loosely like working memory: what is currently available matters.

VISUAL AID

context-window

FEEDBACK

Context is temporary input, not durable training.



Temporary Context Window

Only visible context can influence the next token.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

context window

input context

prompt tokens

response tokens

rag

autoregression

inference

24 / 28

RAG and Retrieval

Open-book AI

Stage: retrieval-inference Priority: high Rubric avg: 4.6/5

DEFINITION

Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's current context before the model generates a response.

DURABLE VS TEMPORARY

RAG usually does not change model weights. Retrieved information is temporary context unless a separate training or storage process occurs.

RELATIONSHIP

RAG depends on the context window. Retrieved notes enter context; attention, hidden states, logits, softmax, and sampling still produce the response.

BRAIN LIMIT

A human can evaluate sources, remember what was read, and reason about trust. The model uses retrieved text as context but does not become aware of it.

CURRENT EXERCISE

open-book-or-learned: Open Book or Learned?

REWRITE NOTES

Risk: RAG is training, permanent memory, guaranteed truth, or consciousness. **Missing:** A plain model answers from learned weights and the current context. A RAG system adds a retrieval step. It searches documents, passages, or records, then inserts relevant snippets into the current context. The model then generates a response using its learned weights plus the retrieved context.

CORE EXPLANATION

A plain model answers from learned weights and the current context. A RAG system adds a retrieval step. It searches documents, passages, or records, then inserts relevant snippets into the current context. The model then generates a response using its learned weights plus the retrieved context.

PROMPT VS RESPONSE

Retrieved material becomes prompt/context tokens. The answer is still generated as response tokens one at a time.

METAPHOR

Open-book exam.

MISCONCEPTION

RAG is training, permanent memory, guaranteed truth, or consciousness.

CHECKPOINT

What does RAG usually do? Correct: Retrieves outside information and places it into the current context. Incorrect: Permanently updates model weights; Guarantees every answer is true; Makes the model aware of documents.

ILLUSTRATION NEEDED

Current visual aid: rag-retrieval. Gold-standard open-book retrieval scene: Prompt, Retriever, Notes, Context, Response, and fixed weights with numbered HTML callouts.

WHERE IT HAPPENS

During inference, before or during response generation.

WHY IT MATTERS

RAG explains many “chat with your documents” systems. The model is not magically knowing private files; the system is retrieving relevant material and showing it to the model.

BRAIN METAPHOR

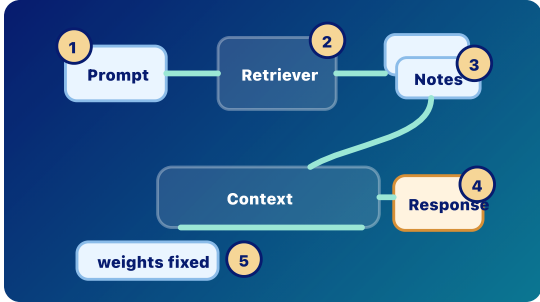
Like checking notes before answering.

VISUAL AID

rag-retrieval

FEEDBACK

Insight unlocked. RAG gives the model temporary evidence in context. Choice notes: Permanently updates model weights: Not quite. RAG is retrieval plus context. It is not training, permanent memory, consciousness, or a truth guarantee. Guarantees every answer is true: Not quite. RAG is retrieval plus context. It is not training, permanent memory, consciousness, or a truth guarantee. Makes the model aware of documents: Not quite. RAG is retrieval plus context. It is not training, permanent memory, consciousness, or a truth guarantee.



Open-Book Retrieval

Retrieval plus context, not training

Retrieved notes enter the context before response tokens are generated.

- 1 **Ask** The user prompt starts the run.
- 2 **Retrieve** A search system finds relevant outside material.
- 3 **Add to context** Retrieved notes become temporary context tokens.
- 4 **Generate** The model still generates response tokens one at a time.
- 5 **Weights stay fixed** RAG does not normally update model weights.

Key takeaway: RAG is retrieval plus context, not training.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 5/5	Exercise 4/5	Mobile 4/5

rag retrieval grounding context window input context prompt tokens response tokens inference fine-tuning training hallucination

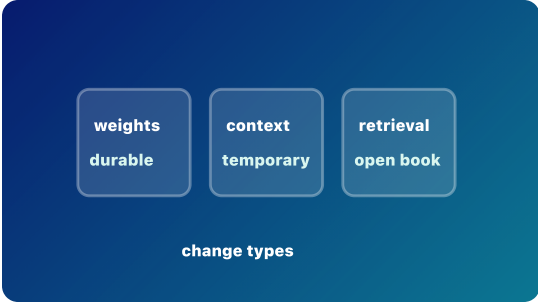
How AI Learns

Durable learning versus temporary steering

Stage: risk/policy

Priority: high

Rubric avg: 3.9/5

DEFINITION AI systems can improve or behave differently through durable training, targeted fine-tuning, retrieval, or temporary in-context steering.	CORE EXPLANATION AI systems can improve or behave differently through durable training, targeted fine-tuning, retrieval, or temporary in-context steering.	WHERE IT HAPPENS These processes happen at different times: before deployment, between deployments, or inside one current run.	 Learning Modes Durable training, retrieval, and temporary steering change different things.
DURABLE VS TEMPORARY Not listed in current app data.	PROMPT VS RESPONSE This card uses the prompt/response path to separate practical risk from myths about secret learning or agency.	WHY IT MATTERS Calling every behavior change learning creates confusion about privacy, memory, and institutional risk.	
RELATIONSHIP Pretraining and fine-tuning can durably change weights; retrieval and examples in the prompt change only what the model can see during inference.	METAPHOR Revising the textbook versus bringing notes into an open-book exam.	BRAIN METAPHOR The word learning is useful when prior experience changes later behavior.	
BRAIN LIMIT In-context examples are not personal learning; they steer the current run without normally updating weights.	MISCONCEPTION Calling every behavior change learning can blur training, retrieval, and in-context steering.	VISUAL AID ai-learns	
CURRENT EXERCISE None rendered in Journey; no direct Play mapping.	CHECKPOINT Which process normally changes only the current context, not weights? Correct: Retrieval-augmented generation. Incorrect: Pretraining; Fine-tuning.	FEEDBACK RAG places retrieved material into the current context. It does not normally train the base model.	
REWRITE NOTES Risk: Calling every behavior change learning can blur training, retrieval, and in-context steering. Missing: Separate durable training, retrieval, temporary instructions, and deployed online learning in one table.	ILLUSTRATION NEEDED Current visual aid: ai-learns. Matrix of durable weight change, temporary context steering, and retrieval.		

Rubric Scores

Accuracy 4/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 3/5

trainingpretrainingfine-tuninginferenceragin-context learning

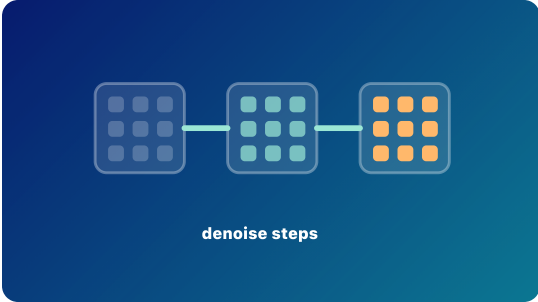
Diffusion vs Autoregression

Denoising is a different generation pattern

Stage: architecture

Priority: medium

Rubric avg: 3.6/5

DEFINITION Diffusion generation usually starts from noise and denoises step by step.	CORE EXPLANATION Diffusion generation usually starts from noise and denoises step by step.	WHERE IT HAPPENS Diffusion is common in image generation and other media workflows, not the standard loop for text LLM generation.	 denoise steps Denoise, Not Append Diffusion refines noise step by step instead of generating text token by token.
DURABLE VS TEMPORARY Not listed in current app data.	PROMPT VS RESPONSE This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.	WHY IT MATTERS The comparison prevents learners from treating all generative AI as the same mechanism.	
RELATIONSHIP Autoregressive LLMs build text token by token; diffusion models refine noise into an image or other output.	METAPHOR Writing the next word versus developing a photograph out of static.	BRAIN METAPHOR Both systems can feel imaginative because they produce new outputs.	
BRAIN LIMIT The mechanisms differ: diffusion denoises a representation; autoregressive text predicts and appends tokens.	MISCONCEPTION Learners may overgeneralize diffusion mechanics to text LLM generation.	VISUAL AID diffusion	
CURRENT EXERCISE None rendered in Journey; no direct Play mapping.	CHECKPOINT What does a diffusion model usually start with? Correct: Noise. Incorrect: A completed paragraph; A spreadsheet.	FEEDBACK Diffusion models learn to denoise toward useful outputs.	
REWRITE NOTES Risk: Learners may overgeneralize diffusion mechanics to text LLM generation. Missing: Show a side-by-side: denoise image representation versus append text tokens.	ILLUSTRATION NEEDED Current visual aid: diffusion. Denoising sequence beside a next-token text loop.		

Rubric Scores				
Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 2/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5
diffusion	autoregression			

Multimodal AI

Multiple media types together

Stage: architecture

Priority: medium

Rubric avg: 3.7/5

DEFINITION

Multimodal AI can represent or process multiple media types, such as text, images, audio, or video.

DURABLE VS TEMPORARY

Not listed in current app data.

RELATIONSHIP

Different modalities may use connected encoders, shared vector spaces, or coordinated model components.

BRAIN LIMIT

Machine modalities are engineered representations, not human sensory experience.

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

REWRITE NOTES

Risk: Human sensory metaphors can make engineered media representations sound more human-like than they are. **Missing:** Add an example of image input, text output, and the representation bridge between them.

CORE EXPLANATION

Multimodal AI can represent or process multiple media types, such as text, images, audio, or video.

PROMPT VS RESPONSE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

METAPHOR

A transit hub where different kinds of information can transfer lines.

MISCONCEPTION

Human sensory metaphors can make engineered media representations sound more human-like than they are.

CHECKPOINT

What does multimodal mean? Correct: Multiple media types. Incorrect: A larger context window only; A model with feelings.

ILLUSTRATION NEEDED

Current visual aid: multimodal. Text, image, audio, and video paths meeting in a shared representation hub.

WHERE IT HAPPENS

It appears when systems connect encoders, decoders, or shared representations across media types.

WHY IT MATTERS

It explains how a system can read an image, answer in text, or combine speech and documents.

BRAIN METAPHOR

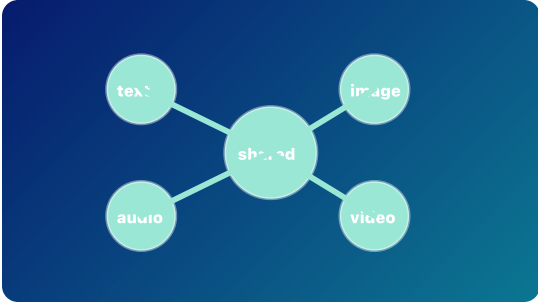
People combine sight, speech, and text naturally, so the metaphor helps.

VISUAL AID

multimodal

FEEDBACK

Multimodal systems work across media types, often through learned representations.



Shared Media Hub
Different media types can connect through learned representations.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

multimodal

embedding

vector

Stage: risk/policy

Priority: high

Rubric avg: 4.3/5

DEFINITION Risk literacy separates practical harms from magical stories about what models are.	CORE EXPLANATION Risk literacy separates practical harms from magical stories about what models are.	WHERE IT HAPPENS It matters whenever people use AI in classrooms, research, administration, health, advising, or IT systems.		
DURABLE VS TEMPORARY Not listed in current app data.	PROMPT VS RESPONSE This card uses the prompt/response path to separate practical risk from myths about secret learning or agency.	WHY IT MATTERS Clear mental models help institutions manage privacy, accuracy, bias, over-trust, and security without inventing magical threats.		
RELATIONSHIP Most real risks come from data, outputs, integrations, incentives, and human over-reliance; ordinary inference does not make the model secretly train itself.	METAPHOR A campus safety map that marks real hazards clearly.	BRAIN METAPHOR Brain metaphors can make risks feel urgent and memorable.		
BRAIN LIMIT Do not infer consciousness, intent, or secret self-training from fluent text. Keep the mechanism visible.	MISCONCEPTION Fluent outputs can invite myths about consciousness, intent, or secret self-training.	VISUAL AID risk		
CURRENT EXERCISE open-book-or-learned: Open Book or Learned?	CHECKPOINT Which is a real institutional risk? Correct: Uploading private data. Incorrect: The model becoming conscious in the chat; Softmax stealing files.	FEEDBACK Risk literacy is part of model literacy.		
REWRITE NOTES Risk: Fluent outputs can invite myths about consciousness, intent, or secret self-training. Missing: Tie each risk to a mechanism: context exposure, tool access, hallucination, bias, or over-trust.	ILLUSTRATION NEEDED Current visual aid: risk. Risk sorting cards linked to mechanisms rather than fear language.			

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 5/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

inference

training

context window

privacy

hallucination

prompt-injection